

# P3Net: Combining the Parameter-less Population Pyramid with a Surrogate Predictor for Neural Architecture Search

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**Abstract—**

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## I. INTRODUCTION

Neural architecture search explores large, discrete spaces of candidate network designs, and assessing any single candidate against the true objective, whether by training it from scratch, through weight sharing, or by querying a benchmark built for this purpose, is the dominant cost of the search [White et al., 2023, Liu et al., 2023]. Treating architecture search and hyperparameter tuning, such as the choice of learning rate or weight decay, as two separate stages, first the architecture under a fixed or default hyperparameter setting, then tuning hyperparameters for the winning architecture, is common practice [Lu et al., 2019, 2020]. Zela et al. [Zela et al., 2018] show it to be unreliable on two counts: architecture and hyperparameters interact, so the best architecture under one hyperparameter setting need not be the best under another; and the common shortcut of ranking architectures after only a few training epochs correlates poorly with their ranking at the full training budget. Separating the two stages compounds this risk twice over. Searching both jointly, rather than sequentially, avoids that compounding, at the cost of a further enlarged search space on top of an already expensive one.

NSGA-Net [Lu et al., 2019] established evolutionary search directly over network architectures, training every candidate to completion. NSGANetV2 [Lu et al., 2020] showed that pairing the same evolutionary engine with a learned accuracy predictor lets most candidates be screened cheaply, reserving full evaluation for a small, promising subset; multi-objective evolutionary search itself remains the field’s most common approach to multi-objective NAS, and surrogate assistance is one of several efficiency techniques built on top of that engine (Related Work).

NSGA-Net’s own experiments document that 60–80% of generated genotypes decode to duplicate architectures as the

number of nodes grows [Lu et al., 2019]. This is evidence that variables in the encoding interact in ways an operator blind to that structure can only exploit by accident, if at all. The cost of duplicated architectures, however, can be neutralised by engineering means external to the search algorithm itself: detecting and skipping genotypes already seen, applied both when building benchmarks (canonicalising isomorphic cells, as in NAS-Bench-101 [Ying et al., 2019]) and on the fly during search (NSGA-Net itself, and P3Net through its own deduplication cache, Proposed Optimizer); this has been standard practice since 2019. This solution is not universal, however. NAS-Bench-201 [Dong and Yang, 2020], a benchmark with its own, separate cell space, does not canonicalise isomorphic architectures: its own authors state they built it “without considering isomorphism,” and of 15,625 raw encodings, only 6,466 (~41%) are topologically unique, an ~59% duplication rate. This same, unmitigated redundancy therefore carries directly into JAHS-Bench-201 [Bansal et al., 2022], which is built on NAS-Bench-201’s cell space and is one of this paper’s two primary benchmarks (Results). What remains open is therefore not the existence of duplicates, but the narrower question of whether a dependency-aware search operator generates fewer of them at proposal time, rather than merely detecting them more cheaply after the fact.

A separate line of work targets exactly this dependency structure between genotype variables: linkage learning evolutionary algorithms such as GOMEA [Thierens and Bosman, 2011] and the Parameter-less Population Pyramid (P3) [Goldman and Punch, 2014], developed largely independently of this NSGA-II-based NAS lineage. They learn which genotype variables interact and protect these groups during crossover, rather than treating the genotype as an unstructured encoding, in principle avoiding the kind of redundant, duplicate-generating search that NSGA-Net’s own numbers document. Duplicates and variable dependencies, however, are two different phenomena that need not go together. Duplicates arise at the decoding step: different genotypes can describe the exact same

architecture, regardless of how the variables affect fitness. Variable dependencies are something else: which genotype variables jointly determine the objective. Such an operator learns exactly these dependencies, not duplicates. Whether it therefore also happens to generate fewer duplicates depends on whether the two phenomena coincide in practice: whether the same variable groups responsible for decoding redundancy are also the groups relevant to fitness. That is exactly what Results’ duplication-rate diagnostic checks.

P3Net combines two elements: a dependency-aware search engine and a relative, linkage-aware surrogate. The engine is P3, which replaces NSGA-II, so that crossover follows a linkage tree learned from the population rather than acting on individual variables. The surrogate, rather than regressing a candidate’s absolute accuracy from its full encoding, predicts the change in accuracy relative to a parent, restricted to the one linkage subset that was modified, in the spirit of eLyMPuS, a variant of LyMPuS [Przewoźniczek et al., 2026] that predicts relative fitness change along a linkage structure discovered incrementally rather than assumed known in advance (Related Work elaborates). Both the search engine and the surrogate are extended to the joint setting, where the genotype includes training hyperparameters such as learning rate and weight decay alongside architectural choices.

Central question and contribution: under a fixed, limited full evaluation budget, does P3’s dependency aware search, paired with a linkage aware, relative surrogate, produce candidates with a better accuracy/computational-cost trade-off than NSGA-II’s crowding distance selection paired with an absolute regressor surrogate, as in NSGANetV2? We test this in the joint architecture and hyperparameter setting. Matched ablations separate the two levers, search engine and surrogate, as far as the design permits; the relative, linkage aware surrogate is only defined given a linkage tree, so it cannot itself be paired with NSGA-II, and Results details this limit of the ablation design. We answer the question through P3Net: a linkage learning evolutionary algorithm paired with a relative, linkage aware surrogate for joint architecture and hyperparameter search, on two benchmarks, JAHS-Bench-201 and NAS-HPO-Bench-II, under equal full evaluation budgets. We evaluate it via a controlled comparison against NSGANetV2, these matched ablations, and SH-EMOA and MO-BOHB, the baselines established for this exact joint setting on these same two benchmarks [Guerrero-Viu et al., 2021]. The closest prior work already combines a surrogate assisted linkage learning algorithm with network search, on an architecture only space [Bartnik, 2026]; Related Work details how P3Net’s linkage aware relative surrogate, its direct comparison against NSGA-II/NSGANetV2, and its joint architecture and hyperparameter scope differ from that baseline.

## II. RELATED WORK

Evolutionary architecture search was pioneered by NSGA-Net [Lu et al., 2019], which employs NSGA-II as the search engine for convolutional network architectures, with selection based on nondominated sorting and crowding distance. Every

candidate in this approach undergoes full training, and the only mechanism accelerating convergence is the exploitation of search history through a Bayesian network. Notably, the authors document substantial redundancy from genotype to phenotype, reporting that 60–80% of generated genotypes correspond to duplicate architectures as the number of nodes grows, which indicates a nontrivial, nonlinear dependency structure hidden within the architecture encoding.

A year later, NSGANetV2 [Lu et al., 2020] introduced an accuracy predictor refined online during the search, and replaced full training from scratch with fine tuning of weights inherited from a supernet, reducing the number of fully evaluated architectures to 350 (Table 2 of the original study). This substitution trades training cost for a documented risk of rank disagreement between weight shared and independently trained accuracy [Yu et al., 2020].

This scheme continues a pattern White et al. [White et al., 2023] identify as the most common approach to multi-objective NAS: evolving architectures with a multi-objective evolutionary algorithm, for which NSGANetV2 itself is cited as an example. Liu et al. [Liu et al., 2023] list pairing that engine with a learned surrogate as one of several recognised techniques for reducing evaluation cost, alongside weight sharing and zero-cost proxies; neither survey singles out this combination as the field’s one dominant paradigm.

Further development in this direction has moved toward predictors based on ranking rather than regression, such as RankNet in MoSegNAS [Lu et al., 2022], which extends the NSGANetV2 approach to semantic segmentation. A separate strand critically reassesses zero cost proxies, whose documented weakness is unreliable discrimination among top ranked architectures.

In parallel, a separate family of evolutionary algorithms has developed that explicitly models structural dependencies between genotype variables, namely GOMEA [Thierens and Bosman, 2011] and the Parameter-less Population Pyramid (P3) [Goldman and Punch, 2014]. These algorithms build a linkage tree from statistical dependencies observed in the population and use it to guide crossover, protecting detected groups of dependent variables from being disrupted. P3 additionally counteracts the premature convergence typical of generational models: unlike NSGA-II, it does not discard previously found good solutions while introducing new diversity. Structurally, P3 replaces a single fixed-size population with an ordered pyramid of populations of growing size (Figure 1), adding a new level only once existing levels stop yielding improved solutions. This property gives the algorithm its name, parameter-less: population size is not chosen in advance. Building and promoting a single new solution up the pyramid canonically requires a real fitness evaluation at every accepted local improvement, not only at the end of the process; this detail matters directly for budget accounting under P3Net’s small evaluation budgets (Results returns to it when discussing the surrogate-free P3 ablation).

It is worth noting that the population level, statistically learned linkage described above should not be confused with

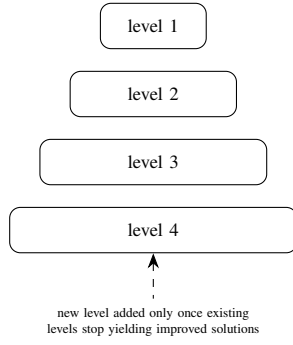


Fig. 1. P3’s population pyramid: each level holds a population of growing size; a new level is added only when the levels below it stop producing improved solutions, so the pyramid’s height is not fixed in advance.

gray box optimization in the narrower sense established by Whitley et al. [Whitley et al., 2016]. That notion is used within the GOMEA line of work, including its real-valued extension RV-GOMEA [Andreadis et al., 2024] and the joint GOMEA library [Bouter and Bosman, 2023]: there, the optimizer is given explicit access to the subfunctions composing the objective and exploits this access to perform cheap, partial fitness reevaluations after a localized variable change. The present work instead operates in the black box regime: NAS validation accuracy cannot be decomposed into locally recomputable subfunctions, so P3 here serves purely as a search engine aware of the genotype’s dependency structure. Cost reduction instead comes from the learned surrogate discussed next.

As with NSGA-Net and NSGANetV2, the cost of full evaluation remains the central bottleneck for linkage learning evolutionary algorithms as well: each optimal mixing step proposes a partial variable change whose effect on the objective must, under this black box regime, be verified through a full fitness evaluation. Pairing this search engine with a cheap learned surrogate that screens candidates before committing to a full evaluation is therefore a natural extension.

Combining linkage learning algorithms with a learned surrogate has already been explored, though not in the NAS domain. CS-GOMEA [Dushatskiy et al., 2019] integrates GOMEA with a convolutional neural network acting as a surrogate, evaluated on synthetic combinatorial problems (Onemax, Trap functions, NK landscapes, HIFF); the authors explicitly identify the surrogate’s failure to exploit linkage tree information in its own construction as an open research direction. Dushatskiy et al. [Dushatskiy et al., 2021] integrate a surrogate directly with a P3 variant, the first such documented case, testing the approach on an expensive, real world partition based ensemble learning problem; notably, an SVR model outperformed a neural predictor in their experiments. The most recent work in this line, LyMPuS [Przewoźniczek et al., 2026], introduces a surrogate integrated directly with the linkage discovery mechanism and verifies it experimentally in combination with P3, again on classical combinatorial benchmarks such as NK landscapes, Ising Spin Glasses, and Max3Sat.

P3Net’s surrogate builds specifically on eLyMPuS, the

“empirical” variant of LyMPuS introduced in the same work. eLyMPuS replaces LyMPuS’s assumption of a known true dependency structure with a dependency structure discovered incrementally, and possibly incompletely, exactly the black-box regime P3Net operates in. Mechanistically, however,  $\hat{\delta}_F$  regresses a continuous fitness difference rather than eLyMPuS’s discrete better/worse/ambiguous comparison, which places its computational mechanism closer to CS-GOMEA’s regression-based relative surrogate [Dushatskiy et al., 2019]. The debt to eLyMPuS is therefore philosophical, not mechanistic: both are relative, linkage-aware estimators over an incompletely discovered structure, and P3Net accordingly forgoes eLyMPuS’s monotonicity-conditional recovery guarantee. Proposed Optimizer details the adaptation.

Among applications of linkage learning algorithms directly to NAS that forgo a learned surrogate entirely, the only identified work is Tran et al. [Tran et al., 2023], combining GOMEA with Synaptic Flow, a training free metric computed analytically that requires no learning from prior evaluation outcomes. This constitutes a fundamentally different cost reduction mechanism than a learned surrogate, and, as discussed below, is distinct from Bartnik’s [Bartnik, 2026] surrogate assisted linkage learning applied to NAS.

Independently of linkage learning approaches, a narrower body of work has addressed the joint optimisation of network architecture and training hyperparameters, rather than treating hyperparameter tuning as a separate step following architecture search [Zela et al., 2018]. Standardised benchmarks for this setting have since been introduced, namely JAHS-Bench-201 [Bansal et al., 2022], a surrogate benchmark over a joint search space combining categorical and continuous variables, and NAS-HPO-Bench-II [Hirose et al., 2021], a fixed grid lookup table over combinations of architecture and hyperparameters. Such benchmarks exist for the same reason NAS-Bench-101 does [Ying et al., 2019]: precomputing or approximating the objective once lets many search runs be compared on identical, reproducible ground truth, rather than repeating live training for every candidate and every compared method. This trade-off, between an exact, enumerable ground truth and a learned, continuous approximation, is discussed further below.

A systematic comparison of solvers on this setting [Guerrero-Viu et al., 2021] evaluates standard baselines such as SH-EMOA (a multi objective evolutionary algorithm built on SMS-EMOA), Multi-Objective Bayesian Optimization Hyperband (MO-BOHB), and random search; none of the methods compared in this line of work is a linkage learning evolutionary algorithm. P3Net adopts SH-EMOA and MO-BOHB as additional baselines directly from this comparison (Results): unlike NSGANetV2, which is imported from architecture only NAS and extended here to the joint setting, SH-EMOA and MO-BOHB were established specifically for joint architecture and hyperparameter search on these same two benchmarks.

Across this joint NAS+HPO literature, the hyperparameter half of the search space is consistently the training pro-

cedure, not the network’s structural capacity. JAHS-Bench-201 [Bansal et al., 2022] searches learning rate, weight decay, activation function, and data augmentation as hyperparameters; the number of cells and the channel width are instead treated as a separate fidelity axis, alongside training epochs and input resolution, distinct from both the architecture and hyperparameter dimensions. NAS-HPO-Bench-II [Hirose et al., 2021] varies only learning rate and batch size as hyperparameters, on top of a fixed cell topology search space; training duration is handled through a separate surrogate rather than as a searched variable.

Zela et al. [Zela et al., 2018] likewise jointly optimise the categorical meta-parameters of a multi-branch ResNet architecture: the number of residual blocks and branches per block, and the per-block widening factor. These are searched together with seven continuous training hyperparameters: initial learning rate, batch size, momentum, L2 weight decay, CutOut length, MixUp  $\alpha$ , and ShakeDrop death rate. Training epochs serve only as the resource dimension of Hyperband here, not as a decision variable.

P3Net follows the same convention (Introduction). Bartnik [Bartnik, 2026] sidesteps this design question entirely, since her genotype has no hyperparameter dimension at all: training procedure is fixed by the NAS-Bench-201 lookup table, and her own discussion acknowledges that the resulting conclusions describe that closed search space rather than NAS in general, a limitation that directly motivates evaluating on genuinely joint benchmarks such as those above.

The joint NAS+HPO literature surveyed above also differs in the source of its ground truth. Zela et al. [Zela et al., 2018] optimise the joint setting through BOHB on real training runs. NAS-HPO-Bench-II [Hirose et al., 2021] answers every query through a fixed lookup table of measured values, while JAHS-Bench-201 [Bansal et al., 2022] answers through a learned surrogate; neither trains anything at evaluation time. Benchmarks of the first kind provide a fixed, exhaustively enumerable lookup table, over which an exact oracle Pareto front is therefore computable; we call these *Category 1*. Benchmarks of the second kind answer every query through a learned, continuous surrogate over the search space and admit no enumerable oracle front; we call these *Category 2*. In this terminology, NAS-HPO-Bench-II is Category 1 and JAHS-Bench-201 is Category 2.

Classifying NAS-HPO-Bench-II as Category 1 assumes that P3Net queries only its exhaustively tabulated range: up to 12 training epochs, with three recorded seeds per entry. The benchmark itself also exposes a separate, learned surrogate (a GIN network combined with an MLP) that extrapolates results to 200 epochs. If that extrapolation were what constituted the full-evaluation level  $r_K$ , it would undermine both this classification and the exact-oracle-front status. It does not: P3Net’s full-evaluation level  $r_K$  for this benchmark is fixed at the tabulated range only, and the 200-epoch surrogate extrapolation is never queried by the search loop. The Category 1 classification and the exact-oracle-front status therefore both hold.

P3Net’s primary benchmarks are these same two, chosen for the reasons already established in this line of work and in the Category 1 NAS literature more broadly: a deterministic objective, a computable oracle front, and a bounded evaluation budget.

JAHS-Bench-201’s own surrogate should not be conflated with the learned surrogate that P3Net, CS-GOMEA [Dushatskiy et al., 2019], Dushatskiy et al. [Dushatskiy et al., 2021], and LyMPuS [Przewoźniczek et al., 2026] build during the search itself. JAHS-Bench-201’s surrogate stands in for the objective function once, offline, before any search begins. The surrogate used by these linkage learning methods works differently: it is fitted incrementally during the search, from candidates already evaluated, purely to decide which further candidates are worth a full query. These are two independent design choices at two different levels; P3Net’s contribution concerns only the second, the search-time surrogate, layered on top of whichever substrate answers the full evaluation.

Among works pairing such a search-time predictor with a linkage learning algorithm, the closest to the present work is Bartnik [Bartnik, 2026]. She adapts SA-P3-GOMEA, a surrogate assisted linkage learning algorithm, to multi objective NAS on NAS-Bench-201, optimising validation accuracy against measured GPU energy consumption. The surrogate used there is an absolute regressor over the flat genotype (deterministic or probabilistic variants of SVR, MLP, random forest, or gradient boosting). The compared baselines are MO-LS and surrogate free MO-P3-GOMEA, not NSGA-II or NSGANetV2. The search space is the closed NAS-Bench-201 benchmark restricted to architecture only, with no training hyperparameters included in the genotype.

Stepping back from Bartnik’s [Bartnik, 2026] specific choices, the broader survey literature confirms this algorithm family’s marginal position: three independent systematic surveys of the evolutionary NAS literature [Liu et al., 2023, White et al., 2023, Özçelik and Efe, 2026] do not list GOMEA, P3, or linkage learning more broadly as a distinct method category, indicating that this algorithm family remains outside the mainstream of the field. The combination of a linkage learning EA with a learned surrogate has already been empirically validated, and, as described above, Bartnik’s [Bartnik, 2026] work is the first to extend it to NAS. Four aspects remain jointly unaddressed, however: a linkage aware, relative surrogate rather than an absolute regressor over the flat genotype; NSGA-II/NSGANetV2 as a directly compared baseline; the joint search over architecture and hyperparameters, to which, to our knowledge, no linkage learning evolutionary algorithm has previously been applied; and the effect of linkage awareness on the genotype duplication rate in a graph-cell search space. Even Bartnik [Bartnik, 2026], the only prior work in this line operating on such a space (NAS-Bench-201, hence susceptible to the same isomorphism as JAHS-Bench-201, discussed in the Introduction), does not report this rate as a separate diagnostic.

### III. PROBLEM FORMULATION

#### Formal definition

Let  $[n] \triangleq \{1, \dots, n\}$ . The search space is defined as the Cartesian product

$$\Lambda = \Lambda_1 \times \Lambda_2 \times \dots \times \Lambda_n,$$

where each  $\Lambda_i$  is a finite set of operations (e.g. CONV3×3, SKIP, NONE). A genotype  $x = (x_1, \dots, x_n) \in \Lambda$  assigns one operation to each edge of the cell graph. Under the joint architecture and hyperparameter search setting targeted in the Results section,  $\Lambda$  is extended with continuous coordinates for training hyperparameters (e.g. learning rate, weight decay), giving  $\Lambda = \Lambda_1 \times \dots \times \Lambda_n \times \Theta$ , where  $\Theta \subset \mathbb{R}^m$  collects these continuous coordinates. Two resolutions of how P3’s block wise linkage tree crossover, defined over discrete groups, is to handle  $\Theta$  remain open, and both are compatible with the formulation below: (i) discretising each coordinate of  $\Theta$  into a finite set, absorbing it into the discrete product above without changing the crossover operator, or (ii) extending the linkage tree and optimal mixing to real valued mixing on  $\Theta$  in the spirit of RV-GOMEA [Andreadis et al., 2024], alongside discrete mixing on  $\Lambda_1 \times \dots \times \Lambda_n$ . Which of the two is used is fixed once, in the Proposed Optimizer section; either choice leaves  $D$ ,  $g$ ,  $f_1$ , and  $f_2$  as defined below unchanged.

#### Decoding and validity.

The decoding function  $D: \Lambda \rightarrow \mathcal{N}$  maps a genotype to a neural network. Not every genotype yields a valid architecture (e.g. no path exists between input and output of the cell). The validity constraint is expressed as

$$g(x) \leq 0,$$

where  $g: \Lambda \rightarrow \mathbb{R}$  is a structural function, boolean in the cases used in this work (e.g.  $g(x) \in \{-1, +1\}$  encoding whether an input–output path exists), stated in real-valued form to accommodate a graded infeasibility measure should a benchmark define one;  $x$  is *valid* if and only if  $g(x) \leq 0$ . The set of valid genotypes is denoted  $\Lambda^* = \{x \in \Lambda : g(x) \leq 0\}$ .

#### Objective functions.

Each genotype  $x \in \Lambda^*$  is assessed by two functions:

$$f_1(x) = \text{validation error of network } D(x) \quad (\text{minimised}), \quad (1)$$

$$f_2(x) = \phi(D(x)) \quad (\text{computational proxy, minimised}), \quad (2)$$

where  $\phi$  denotes the chosen measure of computational cost (parameter count or FLOPs), computed analytically for a fixed input resolution, held at the resolution setting fixed by the highest fidelity level  $r_K$  (Multi fidelity evaluation, below), so that  $f_2$  does not vary with the fidelity level at which  $f_1$  happens to be queried, even on benchmarks such as JAHS-Bench-201 where resolution is itself one of the fidelity axes.

#### Multi objective formulation.

We seek the set of nondominated solutions (Pareto front):

$$\begin{aligned} \mathcal{P}^* = \{ & x \in \Lambda^* : \nexists x' \in \Lambda^*, \\ & f_1(x') \leq f_1(x) \wedge f_2(x') \leq f_2(x) \wedge \\ & \mathbf{f}(x') \neq \mathbf{f}(x) \}, \end{aligned}$$

where  $\mathbf{f}(x) = (f_1(x), f_2(x))$ .

#### Multi fidelity evaluation.

Evaluating  $f_1(x)$  at full fidelity is expensive. We introduce a fidelity ladder  $r_1 < r_2 < \dots < r_K$ , ordered by increasing training epochs; where a benchmark exposes further resource axes affecting evaluation cost or the computational proxy (e.g. input resolution), each level  $r_k$  additionally fixes a corresponding setting on those axes, so  $r_k$  denotes a full resource configuration rather than an epoch count alone. On benchmarks with a single resource axis, this reduces to  $r_k$  denoting the number of training epochs only. The objective at level  $r_k$  is  $f_1^{(k)}(x)$ ; the value at the highest level  $f_1^{(K)}(x)$  is the target quality measure. For model learning and selection, methods that exploit the fidelity ladder use  $f_1^{(k)}(x)$  at the lowest level that correlates sufficiently with  $f_1^{(K)}(x)$ , preventing bias introduced by mixing fidelity levels; P3Net itself does not exploit this ladder, operating exclusively at  $r_K$  (Proposed Optimizer). None of the methods compared in Results (Baselines) exploits it either; it is defined here as general, substrate-level machinery, useful should a future multi-fidelity baseline (e.g. a Hyperband-based method) be added to the comparison, rather than as an active component of the present experiment.

#### Evaluation noise.

Where the substrate answering  $f_1$  carries genuine stochasticity, for instance live training or a benchmark’s own recorded repeats,  $f_1^{(k)}(x)$  is treated as a random variable: each configuration is evaluated with  $s$  distinct seeds, or with samples drawn from the substrate’s recorded repeats, and the median across them is used for selection and reporting. Deterministic substrates, such as a point estimate surrogate, require no such averaging. Both of P3Net’s primary benchmarks (Results) answer queries deterministically: a point-estimate surrogate for JAHS-Bench-201, a fixed lookup table for NAS-HPO-Bench-II within its exhaustively tabulated range (caveat on which fidelity level this refers to discussed in Related Work). So  $s = 1$  throughout this work and this averaging step is not exercised; it is retained as general substrate-level machinery for future work on stochastic substrates such as live training.  $s$  should not be confused with  $R$ , the number of independent search runs per method (Results):  $R$  remains meaningful regardless of substrate determinism, since randomness across runs comes from initialization and stochastic search operators, not from repeated queries to the same configuration.

#### Surrogate model.

A full evaluation of  $f_1(x)$  queries the true objective for  $D(x)$ ; the concrete substrate for this query, whether live training, a benchmark lookup, or a benchmark’s own surrogate, is instantiated in the Results section. After collecting  $t$  full evaluations we have the observation dataset

$$\mathcal{H}_t = \{(x^{(i)}, \mathbf{f}(x^{(i)}))\}_{i=1}^t.$$

Rather than regressing  $f_1$  directly from an encoded genotype, as in NSGANetV2, the surrogate used by P3Net is *relative* and *linkage aware*: for a parent  $x \in \Lambda^*$  and a candidate  $x'$  differing

from  $x$  only on the coordinates of a linkage subset  $F \subseteq [n]$  identified by the current dependency model (i.e.  $x'_i = x_i$  for all  $i \notin F$ ), the surrogate predicts the effect of that local change relative to the parent,

$$\hat{\delta}_F: \Lambda^* \times \Lambda^* \rightarrow \mathbb{R}, \quad \hat{\delta}_F(x, x') \approx f_1(x) - f_1(x'),$$

trained on pairwise fitness differences derived from  $\mathcal{H}_t$ . The primary cost of search is measured in calls to  $f_1$  (full evaluations), not in calls to  $\hat{\delta}_F$ . This is what distinguishes P3Net from NSGANetV2 already at the level of problem formulation: P3Net scores candidates relative to their parent using the structure discovered by P3, whereas NSGANetV2 scores each candidate in absolute terms from its encoding alone. The construction and update of  $\hat{\delta}_F$ , and its use within the P3Net search loop, are described in the Proposed Optimizer section.

#### IV. PROPOSED OPTIMIZER

##### *Representation and linkage tree.*

A linkage tree is built from the current population, reflecting statistical dependencies between genotype variables: the per-edge operation choices of Problem Formulation, extended below to include the discretised  $\Theta$  coordinates once the joint setting is in scope. Following the standard P3 procedure, the tree is constructed by agglomerative, UPGMA style hierarchical clustering over a normalised mutual information based dependency measure between genotype variables, yielding the nested family of variable subsets used by the optimal mixing operator (Figure 3); this construction is applied to the NAS genotype unchanged from its original combinatorial optimisation form. P3Net adopts option (i) from Problem Formulation: each continuous coordinate of  $\Theta$  is discretised into a finite set of bins, fixed once before search (e.g. a log-spaced grid for learning rate), so that the joint genotype  $\Lambda_1 \times \dots \times \Lambda_n \times \Theta$  is entirely categorical and the linkage tree, its mutual-information-based dependency measure, and the optimal mixing sweep described below apply to it exactly as to the architecture-only encoding, with no separate real-valued mixing mechanism. This trades the resolution of the continuous hyperparameter search for keeping the central comparison against NSGA-II/NSGANetV2 attributable to the search engine and surrogate alone, rather than conflating it with an additional, separately novel continuous-mixing extension to P3; extending to option (ii) (real valued mixing in the spirit of RV-GOMEA [Andreadis et al., 2024]) is left as future work.

##### *Crossover.*

Crossover operates on the level of variable groups identified by this tree, rather than individual bits, protecting detected groups of related design decisions from being disrupted.

##### *No mutation operator.*

P3Net, like the wider P3/GOMEA lineage it draws on [Goldman and Punch, 2014, Thierens and Bosman, 2011], has no separate mutation step: all variation is produced by the crossover-guided optimal mixing sweep described below, so population diversity comes entirely from recombination

against the linkage tree rather than from an additional random perturbation operator.

##### *Surrogate.*

The surrogate’s construction in P3Net exploits the same dependency structure that guides the variation operator: the relative, linkage aware estimator  $\hat{\delta}_F$  defined in Problem Formulation, trained on pairwise fitness differences derived from  $\mathcal{H}_t$ , is what P3’s optimal mixing sweep queries at every proposed modification. Mechanistically this regression-based estimator is closer to CS-GOMEA’s relative surrogate [Dushatskiy et al., 2019] than to eLyMPuS’s discrete comparison; the debt to eLyMPuS, a variant of LyMPuS [Przewoźniczek et al., 2026], is philosophical rather than mechanistic. Both predict relative fitness change conditioned on an incompletely discovered linkage structure rather than one assumed known in advance (Related Work elaborates), adapted here to network architecture encoding.

##### *Telescoping construction.*

An estimate of a candidate’s absolute error is obtained compositionally, by telescoping this relative correction back along the sweep to the nearest ancestor with a known, fully evaluated value of  $f_1$ : writing  $x_0, x_1, \dots, x_m = x'$  for the chain of tentatively accepted modifications since that ancestor  $x_0$  (with  $f_1(x_0)$  known), each step contributing  $\hat{\delta}_{F_i}(x_{i-1}, x_i)$  for the linkage subset  $F_i$  modified at step  $i$ ,

$$\hat{f}_1(x_m) = f_1(x_0) - \sum_{i=1}^m \hat{\delta}_{F_i}(x_{i-1}, x_i),$$

which reduces to  $\hat{f}_1(x') = f_1(x) - \hat{\delta}_F(x, x')$  in the single step case  $m = 1$ . The chain depth  $m$  is bounded by  $\kappa$ , introduced below, precisely to limit how far this accumulated surrogate error can propagate before a full evaluation resets it.

##### *Search loop.*

Figure 2 summarises one iteration; the loop proceeds as follows:

- 1) P3 generates a candidate set  $C \subset \Lambda^*$  through crossover guided by the linkage tree; for each parent selected for variation, every linkage subset in the current dependency model is visited in random order, following the standard optimal mixing sweep, and for each subset  $F$  a donor is drawn from the population to propose a modification restricted to  $F$  (Parent, donor, and ancestor provenance, below);
- 2) every proposed modification  $x'$  is scored by the surrogate relative to its immediate predecessor in the sweep,  $\hat{f}_1(x')$ , obtained via  $\hat{\delta}_F$ ;
- 3) a modification is tentatively accepted if  $\hat{\delta}_F$  predicts nonnegative improvement (a zero acceptance threshold by default; Results assesses sensitivity to this threshold jointly with  $\kappa$ ), and the sweep continues from the resulting individual; otherwise the modification is rejected, the individual is left unchanged, and the sweep proceeds to the next linkage subset in the order fixed in step 1;
- 4) once every parent’s sweep has run to completion, or has been curtailed by the chain depth limit  $\kappa$  described

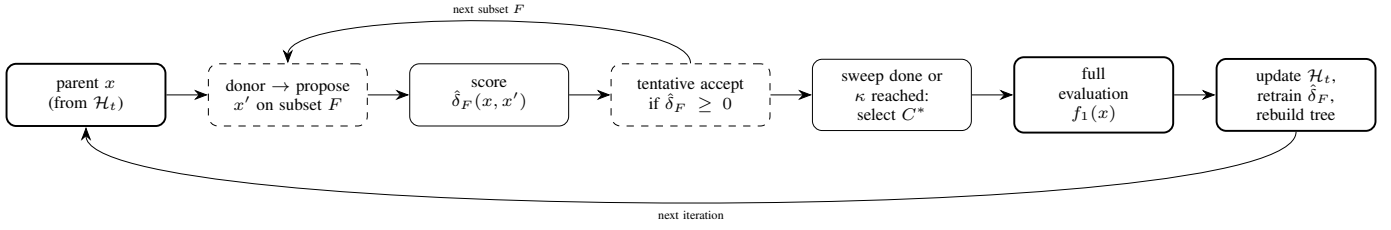


Fig. 2. One P3Net search-loop iteration (Search loop, below). Solid boxes hold individuals with a real, fully evaluated  $f_1$  (members of  $\mathcal{H}_t$ ); dashed boxes hold transient individuals accepted only by the surrogate  $\hat{\delta}_F$ , never yet confirmed by a full evaluation. The upper loop repeats for each linkage subset visited in the sweep, up to the chain-depth limit  $\kappa$  (Chain depth, below); the lower loop begins the next search iteration.

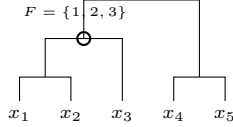


Fig. 3. Illustrative linkage tree over five genotype variables. Leaves are individual coordinates of the joint genotype  $\Lambda_1 \times \dots \times \Lambda_n \times \Theta$ ; each internal node is a candidate linkage subset  $F$  that may be visited during the optimal mixing sweep (step 1, Search loop, below). A concrete subset,  $F = \{1, 2, 3\}$ , is highlighted.

below, a promising subset  $C^* \subseteq C$  is selected among all sweep outcomes as the candidates nondominated in the  $(\hat{f}_1, f_2)$  objective space ( $C^*$  is an iteration-local selection set, distinct from the global front  $\mathcal{P}^*$  of Problem Formulation that the search approximates over time);

- 5) each  $x \in C^*$  undergoes a full evaluation, yielding the true value  $f_1(x)$ ;
- 6)  $\mathcal{H}_t$  is updated,  $\hat{\delta}_F$  is retrained, and the linkage tree is rebuilt as new data arrives, evolving alongside the population; within a single iteration, the same tree built at the start of that iteration is used for every parent's sweep in step 1, and is not rebuilt again until the next pass through step 6.

#### Parent, donor, and ancestor provenance.

Parents for step 1, and the ancestor  $x_0$  used in the telescoping construction (above), are always drawn from individuals that have completed steps 5–6, i.e. elements of  $\mathcal{H}_t$  with a genuinely known  $f_1$ , never from the transient, surrogate-only individuals produced mid-sweep; the telescoping construction therefore never bottoms out on a surrogate estimate. (TODO: the donor drawn within step 1 is not yet subject to this same stated restriction; clarify whether donors, like parents and the ancestor  $x_0$ , must also come from  $\mathcal{H}_t$ , or may also be transient, surrogate-only individuals from within the same sweep.)

#### Fidelity scope.

P3Net operates exclusively at full fidelity  $r_K$ : the lower-fidelity apparatus introduced in Problem Formulation is substrate-level machinery, instantiated where applicable in Results, but is not consumed by the search loop above. Training the surrogate on lower-fidelity observations as well is a design alternative discussed in Conclusions.

#### Constraint handling.

Candidates violating the constraint  $g(x) \leq 0$  are rejected prior to surrogate scoring (i.e. before step 2 of the loop above).

#### Deduplication.

Duplicates already present in  $\mathcal{H}_t$  skip reevaluation. This is applied identically to every compared method as part of the shared evaluation harness (Results, Fairness controls), so the saving is not specific to P3Net; it is a safeguard of substantial practical relevance given the genotype-to-phenotype redundancy documented in NSGA-Net (Introduction) [Lu et al., 2019]. The computational cost  $f_2$  itself is computed analytically, without surrogate involvement (Problem Formulation), so selection of candidates for full evaluation relies on non-dominance in the  $(\hat{f}_1, f_2)$  space.

#### Chain depth ( $\kappa$ ).

To bound the accumulation of surrogate error along a sweep, a maximum chain depth  $\kappa$  of tentatively accepted modifications based only on the surrogate is imposed: once  $\kappa$  modifications have been accepted since the last full evaluation along a given sweep, the current individual is forced into  $C^*$  regardless of its predicted  $\hat{f}_1$ .  $\kappa$  is initialised at eLyMPuS's own  $2\lceil\log_2(n)\rceil$ -step bound, as a heuristic default, not as a derivation from the monotonicity guarantee that bound relies on, which does not apply here. Here  $n$  is the dimensionality of the full extended, discretised genotype  $\Lambda_1 \times \dots \times \Lambda_n \times \Theta$ , matching the scope of the linkage tree itself, not the architecture-only count of Problem Formulation; its sensitivity, jointly with that of the step-3 acceptance threshold (also a heuristic default, fixed at zero above), is assessed empirically rather than fixed by assumption (Results).

#### Design rationale (adapted from eLyMPuS).

Three further design choices follow from adapting the eLyMPuS mechanism to this setting:

- Candidate generation follows the full optimal mixing sweep described in step 1, rather than a single modification per candidate, so that P3Net remains consistent with the variation operator on which P3's own performance guarantees rely.
- eLyMPuS's guarantee of recovering a missing dependency within  $2\lceil\log_2(n)\rceil$  steps relies on a monotonicity assumption about the underlying fitness landscape that is not expected to hold for neural network accuracy;  $\hat{\delta}_F$  is therefore treated as a learned heuristic estimator without

that guarantee, and its reliability is instead assessed empirically through the surrogate quality analysis reported in the Results section.

- Nondomination in step 4 is restricted to candidates produced within the same search iteration, i.e. propagated from ancestors evaluated against the same snapshot of  $\mathcal{H}_t$ , so that compared  $\hat{f}_1$  estimates share a common reference point rather than being pooled across iterations with a potentially drifting surrogate.

## V. RESULTS

This section reports surrogate quality, convergence under fixed evaluation budgets, Pareto front and hypervolume comparisons, and a fixed-budget summary table (median, IQR, corrected  $p$  values, effect sizes), in that order. The experimental setup and protocol are detailed first.

### Experimental Setup

The protocol must not allow P3Net to be read as “P3 plus Pareto filtering.” It states explicitly which selection mechanics NSGA-II contributes to the baseline, what P3 replaces (the variation operator and the proposal distribution), and, separately from the choice of search engine, which surrogate variant is used (linkage aware  $\hat{\delta}_F$  vs. an absolute regressor), so that gains can be attributed to the correct component rather than conflated.

#### Benchmark and search space.

P3Net targets the joint architecture and hyperparameter search setting rather than an architecture only space. Rather than resolving the Category 1 vs. Category 2 question by picking a single benchmark, both are reported side by side, deliberately. The primary benchmark is JAHS-Bench-201 [Bansal et al., 2022], a surrogate benchmark over a joint search space of ten dimensions (six categorical architecture edges plus four hyperparameters, including continuous learning rate and weight decay), fourteen when JAHS-Bench-201’s four additional fidelity dimensions are counted alongside it, per the source’s own “14-dimensional search and fidelity space” figure, across three datasets; it queries a learned surrogate over its full mixed categorical/continuous search space without live training, so no enumerable oracle front exists (see Metrics, below). NAS-HPO-Bench-II [Hirose et al., 2021], a fixed grid lookup table over combinations of architecture and hyperparameters, is reported alongside it as the Category 1 counterpart: within its exhaustively tabulated range (up to 12 training epochs; Related Work confirms  $r_K$  is fixed at this level, not the benchmark’s separate 200-epoch surrogate extrapolation), an exact oracle front is enumerable, enabling the Inverted Generational Distance plus (IGD+) on this benchmark specifically. Any result that holds on both benchmarks is reported as the primary finding; a result that holds on only one is reported as specific to that benchmark and not over generalised.

#### Baselines.

NSGANetV2 (NSGA-II with an absolute regressor surrogate) as the primary baseline; NSGANetV2 extended to

TABLE I

THE NINE BASELINES/ABLATIONS (BASELINES, ABOVE), ORGANISED BY SEARCH ENGINE AND SURROGATE TYPE. THE NSGA-II/RELATIVE CELL IS NOT CONSTRUCTIBLE, SINCE  $\hat{\delta}_F$  IS DEFINED ONLY GIVEN A LINKAGE TREE. FIVE FURTHER BASELINES, NAMELY NSGANetV2 WITH CONTINUOUS  $\Theta$ , SH-EMOA, MO-BOHB, RANDOM SEARCH, AND TPE, SIT OUTSIDE THIS GRID ENTIRELY AND ARE NOT ENGINE/SURROGATE ABLATIONS.

|                | No surrogate | Absolute regressor | Relative $\hat{\delta}_F$ |
|----------------|--------------|--------------------|---------------------------|
| <b>NSGA-II</b> | NSGA-Net     | NSGANetV2          | —                         |
| <b>P3</b>      | P3 alone     | P3 + absolute      | P3Net (ours)              |

the joint setting with an unconstrained, real-valued encoding of  $\Theta$  as an additional control baseline isolating the effect of discretisation itself, rather than of the search engine or surrogate (Fairness controls); NSGA-Net (NSGA-II with no surrogate, every candidate fully evaluated) as the engine-only ablation on the NSGA-II side, symmetric to the P3-only ablation below; P3 alone, without any surrogate, as the engine-only ablation on the P3 side; P3 with an absolute regressor surrogate (in the style of NSGANetV2) as an ablation isolating only the surrogate, separating the contribution of the linkage aware design from the contribution of the P3 engine itself; SH-EMOA (a multi objective evolutionary algorithm built on SMS-EMOA) and Multi-Objective Bayesian Optimization Hyperband (MO-BOHB), the baselines established for this exact joint setting on these same two benchmarks [Guerrero-Viu et al., 2021], included because, unlike NSGANetV2, they were never imported from an architecture only setting and require no such extension; random search and the Tree-structured Parzen Estimator (TPE) as sanity baselines. This grid separates the engine and surrogate contributions as far as the design permits, not as a full orthogonal factorial: the relative, linkage aware surrogate  $\hat{\delta}_F$  is defined only given a linkage tree, so an NSGA-II-plus-relative-surrogate cell is not constructible, and any gap attributed to “the surrogate” via the P3-only vs. P3Net comparison should be read against this asymmetry.

Without a surrogate, P3 alone follows the canonical optimal-mixing acceptance rule: every proposed modification within a sweep is gated by a *real* evaluation of  $f_1$ , exactly as step 3 of the P3Net loop (Proposed Optimizer) would be if  $\hat{\delta}_F$  were replaced by  $f_1$  itself, and as in the original P3 algorithm’s own hill-climbing and cross-level mixing steps [Goldman and Punch, 2014]. Under the  $\{50, 100, 200\}$ -evaluation budgets used here (Budgets, below), this is expected to exhaust most or all of the budget within a small number of sweeps: for a joint genotype with dozens of linkage subsets, a single parent’s full sweep alone can approach the smallest tier. This is not a flaw in the ablation but its point: if P3 alone collapses under this budget, that result directly motivates pairing P3 with a cheap surrogate in the first place, which is what the central question (Introduction) actually compares. We report the number of full sweeps completed within budget for this ablation specifically, alongside its hypervolume, so a weak result reads as budget starvation rather than as a failure of the search engine itself.

*Surrogate error accumulation ( $\kappa$  and acceptance threshold).*

The main comparison above fixes  $\kappa$  at its default value, eLyMPuS’s own  $2\lceil\log_2(n)\rceil$ -step bound (Proposed Optimizer), and the step-3 acceptance rule at its default zero threshold ( $\hat{\delta}_F \geq 0$ ). Both are swept jointly in the sensitivity analysis:  $\kappa$  over a small grid (e.g.  $\{1, \lceil\log_2(n)\rceil, 2\lceil\log_2(n)\rceil, \infty\}$ ), and the acceptance threshold over a small margin grid (e.g.  $\{0, \epsilon, 2\epsilon\}$  for a fixed  $\epsilon > 0$ ), with P3Net otherwise unchanged, reporting fixed budget hypervolume and surrogate rank correlation as a function of both.

*Fairness controls.*

Identical genotype encoding, decoding function  $D$ , validity handling, full evaluation budget, seed policy, stopping rule, and deduplication cache (Proposed Optimizer) across all compared methods, so that no method’s apparent efficiency benefits from infrastructure the others lack. In particular, the discretisation of continuous hyperparameter coordinates (Proposed Optimizer) is applied identically to every baseline: NSGA-II/NSGANetV2 search the same discretised  $\Theta$  as P3Net rather than the unconstrained real-valued encoding a joint-setting extension of NSGA-II/NSGANetV2 would otherwise use, so that any gain is attributable to the search engine and surrogate rather than to a difference in genotype resolution. This equalises resolution but is not necessarily neutral: discretisation removes the real-valued crossover on  $\Theta$  that such an extension would otherwise use, while costing P3 nothing, since P3 operates on a categorical genotype regardless. This asymmetry is bounded empirically rather than merely acknowledged: NSGANetV2 in its native continuous encoding is included as an additional control baseline (Baselines, above), so any P3Net advantage can be checked specifically against the discretised-NSGANetV2 comparison, rather than resting on the discretised comparison alone.

*Budgets, seeds, stopping.*

Three full evaluation budget tiers,  $\{50, 100, 200\}$  full evaluations, doubling at each step so convergence trends read on a natural, geometric scale. Each tier is anchored to a figure actually used in the surrounding literature rather than chosen arbitrarily: the middle tier matches JAHS-Bench-201’s own suggested evaluation protocol of approximately 100 full evaluations, with a minimum of 10 seeds [Bansal et al., 2022]; the top tier stays at or below the 350 full evaluations NSGANetV2 itself uses to reach its reported results [Lu et al., 2020], so no tier exceeds the regime in which the primary baseline was originally shown to work well. This also keeps every tier below NAS-HPO-Bench-II’s own convention of 500 trials for benchmarking search algorithms [Hirose et al., 2021], a deliberate choice, not an oversight: the central question concerns a *limited* budget (Introduction), so all three tiers stress-test a regime tighter than either benchmark’s own default, on both benchmarks alike.  $R = 10$  independent runs per method with a fixed, published seed list, matching JAHS-Bench-201’s recommended minimum [Bansal et al., 2022]; identical initialization scheme across methods; a stopping rule covering both the evaluation budget and, if the search space is

small enough to converge prematurely, an exploration collapse criterion analogous to the one Bartnik [Bartnik, 2026] needed to add for spaces at the scale of NAS-Bench-201.

*Metrics.*

Fixed budget hypervolume and, only if the search space admits an enumerable oracle front, IGD+; otherwise hypervolume relative to the best known front is used in its place (*TODO: define this front precisely, e.g. as the union of all points evaluated by any compared method across all runs*). Surrogate quality is reported separately as rank correlation (or pairwise comparison accuracy, if the surrogate is relational) between predicted and true values as a function of  $|\mathcal{H}_t|$ .

*Statistical plan.*

The comparison set is P3Net against each of the other nine arms (Baselines, above), per benchmark and per budget tier, not every pairwise combination among all arms. Paired Wilcoxon signed rank tests are run per comparison in this set, with Holm–Bonferroni correction applied across it, and an effect size (e.g. Cliff’s delta) reported alongside every corrected  $p$  value. This is stricter than the uncorrected comparisons in the closest prior work [Bartnik, 2026], which report no effect size, and the difference is treated as a deliberate methodological improvement, not an oversight to reconcile.

*Diagnostics.*

Genotype duplication rate and archive turnover over the course of the search, motivated directly by the genotype-to-phenotype redundancy documented in NSGA-Net (Introduction) [Lu et al., 2019], measured under the identical joint genotype encoding fixed for every compared method (Fairness controls), so that any difference in duplication rate reflects the search operator rather than a difference in encoding. The deduplication cache (Proposed Optimizer) is applied identically to every compared method as part of the shared evaluation harness (Fairness controls), so the measured duplication rate reflects what each search operator proposes, not an artifact of asymmetric caching. This rate is counted at proposal time, for every candidate a search operator generates, whether or not it is later intercepted by the cache, not at evaluation time, so it remains informative even though the cache itself suppresses re-evaluation of anything it counts. Convergence curves; Pareto front progression; stability across seeds.

*Reproducibility.*

Every cached evaluation is keyed by genotype, experiment type, and a protocol version string covering every setting that affects the recorded value, so that a change in any of them invalidates stale cache entries automatically, following the pattern used by [Bartnik, 2026].

*Findings*

(*TODO: report results here, in the order promised above: surrogate quality (rank correlation vs.  $|\mathcal{H}_t|$ ); convergence under each fixed evaluation budget; Pareto front and hypervolume comparisons across all nine baselines/ablations on both benchmarks; and the fixed-budget summary table (median, IQR, corrected  $p$  values, effect sizes) from the Statistical plan.*)

Include the  $\kappa$ /acceptance-threshold sensitivity analysis and the duplication-rate diagnostic once run.)

## VI. CONCLUSIONS

- P3Net combines two complementary ideas: P3 builds a model of structural dependencies in the genotype, the surrogate eliminates costly evaluations of unpromising candidates.
- Answer to the research question: whether P3’s dependency modelling translates into better surrogate loop efficiency compared to NSGANetV2.
- Limitations: surrogate quality plausibly depends on genotype encoding quality, though this dependency is not isolated by a dedicated ablation in the present design; the linkage tree is built from pairwise mutual information, which may be too coarse a signal to capture higher-order, deeply structured dependencies. JAHS-Bench-201 answers every query through a learned surrogate rather than measured data, so results on it inherit that surrogate’s approximation error relative to real, independent training; since every compared method queries the same surrogate, systematic bias affects them equally, but this does not rule out the error correlating specifically with the structure P3Net exploits. A result that also holds on NAS-HPO-Bench-II, built from measured data, is therefore treated as the primary finding, and a result specific to JAHS-Bench-201 alone as requiring confirmation.
- Future directions: stronger surrogate model (GNN over the cell graph), active learning to decide which candidates to evaluate fully, extension to architecture spaces with variable depth; training the surrogate on lower-fidelity observations  $f_1^{(k)}$ ,  $k < K$ , rather than exclusively on full evaluations at  $r_K$  as in the present work, using the fidelity ladder already introduced in Problem Formulation. This is left out here to keep the search loop’s cost reduction cleanly attributable to the relative surrogate alone, but is a natural next lever once that attribution is established.

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# NOTES: PROBLEM FORMULATION DESIGN LOG

*Working log, not part of the reviewed manuscript body: every architectural decision made while formulating the problem, the surrogate, and the search engine, with the reasoning behind each choice and where it lives in the text above.*

## Problem representation

| Decision                              |        | Specifics                                                                                                                                                                                                                                                            | Rationale / consequence                                                                                                                                                                                                                                                                                                                             | Location                   |
|---------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Architecture type                     | geno-  | Vector $x = (x_1, \dots, x_n) \in \Lambda$ , each $x_i$ from a finite set of operations on a cell-graph edge                                                                                                                                                         | Standard cell-based NAS encoding (as in NSGA-Net)                                                                                                                                                                                                                                                                                                   | Problem Formulation        |
| Joint-search extension                | exten- | $\Lambda \rightarrow \Lambda_1 \times \dots \times \Lambda_n \times \Theta$ , $\Theta \subset \mathbb{R}^m$ for training hyperparameters (LR, weight decay)                                                                                                          | Answers Zela et al.: architecture and hyperparameters interact, so sequential tuning is unreliable                                                                                                                                                                                                                                                  | Problem Formulation        |
| Scope of $\Theta$                     |        | Training procedure only (LR, weight decay); structural capacity (cell count, channel width) stays part of the architecture / fidelity axes, not $\Theta$                                                                                                             | Matches the convention across the joint NAS+HPO literature (JAHS-Bench-201, NAS-HPO-Bench-II, Zela et al.)                                                                                                                                                                                                                                          | Related Work               |
| Handling $\Theta$ 's continuity in P3 |        | <b>Discretisation</b> (option i): each $\Theta$ coordinate is binned into a finite set before linkage learning; no separate mixing mechanism                                                                                                                         | Deliberate trade-off: lower HP-space resolution in exchange for a clean central question (does not conflate testing P3 with testing a novel continuous-mixing extension); RV-GOMEA-style mixing left as future work                                                                                                                                 | Proposed Optimizer         |
| Encoding in base-lines                |        | Same $\Theta$ discretisation imposed on NSGA-II/NSGANetV2 (rather than the unconstrained real-valued encoding a joint-setting extension of them would otherwise use); NSGANetV2 additionally run once more with that unconstrained real-valued encoding as a control | Discretisation is required for the “identical genotype encoding” fairness rule, but is not neutral: it costs NSGA-II/NSGANetV2 the real-valued crossover such an extension would otherwise use, while costing P3 nothing; the continuous- $\Theta$ control run bounds this asymmetry empirically instead of only asserting fairness by construction | Results, Fairness controls |

## Decoding and validity

| Decision             |         | Specifics                                                                                       | Rationale / consequence                         | Location            |
|----------------------|---------|-------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------|
| Decoding             |         | $D : \Lambda \rightarrow \mathcal{N}$ , maps a genotype to a network                            | —                                               | Problem Formulation |
| Validity             |         | Constraint $g(x) \leq 0$ (e.g. no input–output path); $\Lambda^*$ is the set of valid genotypes | Not every genotype decodes to a working network | Problem Formulation |
| Rejecting candidates | invalid | $g(x) > 0$ rejected before surrogate scoring                                                    | Saves surrogate evaluation cost                 | Proposed Optimizer  |

## What is being optimised

| Decision    |  | Specifics                                                                                               | Rationale / consequence                                  | Location            |
|-------------|--|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------|
| Objective 1 |  | $f_1(x)$ = validation error of $D(x)$ , minimised                                                       | Standard predictive quality                              | Problem Formulation |
| Objective 2 |  | $f_2(x) = \phi(D(x))$ = computational cost (parameter count or FLOPs), computed analytically, minimised | Classic two-objective NAS formulation (quality vs. cost) | Problem Formulation |

| Decision                    | Specifics                                                                                                           | Rationale / consequence                                                      | Location            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------|
| Resolution used for $f_2$   | Fixed at the highest fidelity level $r_K$ 's setting, independent of the level at which $f_1$ happens to be queried | $f_2$ must not drift with fidelity level, so candidate costs stay comparable | Problem Formulation |
| Multi-objective formulation | Pareto front $\mathcal{P}^*$ — standard dominance relation over $(f_1, f_2)$                                        | —                                                                            | Problem Formulation |

#### *Budget and evaluation fidelity*

| Decision                   | Specifics                                                                                                                                                                                                                                          | Rationale / consequence                                                                                                                                                                                                                                                                                                               | Location                                 |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Fidelity ladder            | $r_1 < \dots < r_K$ , ordered by epoch count; if a benchmark exposes further resource axes (e.g. resolution), each level also fixes their value                                                                                                    | Defined as general, substrate-level machinery (not tied to any one benchmark); full evaluation of $f_1$ is expensive, so a ladder is the natural vehicle for cheap filtering at lower fidelity, <i>if</i> a method chooses to use one                                                                                                 | Problem Formulation                      |
| P3Net's own fidelity scope | P3Net operates exclusively at full fidelity $r_K$ ; the ladder is <b>not</b> consumed by the search loop — no candidate is ever scored via $f_1^{(k)}$ , $k < K$ ; none of the Results baselines use it either                                     | Keeps the search loop's cost reduction cleanly attributable to the relative surrogate alone, avoiding a third, unablated cost-reduction lever; kept as general infrastructure for a possible future multi-fidelity baseline (e.g. Hyperband-based), not an active component now; training on lower fidelities is noted as future work | Proposed Optimizer; Results; Conclusions |
| Noise handling             | Stochastic substrates (live training, benchmark repeats): $s$ seeds, median used for selection/reporting; deterministic substrates (benchmark surrogate): no averaging. Both of P3Net's chosen benchmarks are deterministic, so $s = 1$ throughout | Selection reliability under a non-deterministic objective, kept general for future stochastic substrates; $s$ (repeats of one query) is distinct from $R$ (independent search runs, Results) — the two should not be conflated                                                                                                        | Problem Formulation                      |
| Cost accounting            | Counted exclusively in calls to $f_1$ (full evaluations), never in calls to $\hat{\delta}_F$                                                                                                                                                       | Consistent with the paper's whole premise: full evaluation cost is the bottleneck                                                                                                                                                                                                                                                     | Problem Formulation                      |

#### *Search engine*

| Decision          | Specifics                                                                                                                                                                                        | Rationale / consequence                                                                                                                                                                                                                                      | Location           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Algorithm         | P3 instead of NSGA-II                                                                                                                                                                            | Tests the hypothesis that dependency awareness beats crowding-distance selection under the same budget                                                                                                                                                       | Proposed Optimizer |
| Pyramid structure | P3 grows an ordered pyramid of populations of increasing size instead of a single fixed-size population, adding a level only once existing levels stop improving — the “parameter-less” property | Explains why the name refers to a pyramid; canonically, promoting one solution up the pyramid costs a real evaluation per accepted local improvement, which is why the surrogate-free P3 ablation is expected to be budget-starved (see Baselines row below) | Related Work       |
| Linkage tree      | Agglomerative UPGMA clustering over normalised mutual information between genotype variables, standard P3 procedure, unmodified for the architecture part                                        | Reuses a validated P3 mechanism unchanged                                                                                                                                                                                                                    | Proposed Optimizer |

| Decision                 | Specifics                                                                                     | Rationale / consequence                                                                                                           | Location           |
|--------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Tree rebuild granularity | Fixed for all sweeps within one iteration; rebuilt only at step 6, at iteration boundaries    | Removes ambiguity about whether the tree could change mid-sweep — it cannot                                                       | Proposed Optimizer |
| Crossover                | Operates on variable groups (linkage subsets), not individual genes                           | Protects discovered groups of related design decisions from disruption                                                            | Proposed Optimizer |
| No mutation operator     | All variation comes from the crossover-guided optimal-mixing sweep; no separate mutation step | Inherited from the P3/GOMEA lineage; diversity relies entirely on recombination against the linkage tree, not random perturbation | Proposed Optimizer |

### Surrogate

| Decision                            | Specifics                                                                                                                                                                                                                                         | Rationale / consequence                                                                                                                                            | Location                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Surrogate type                      | <b>Relative and linkage aware:</b> $\hat{\delta}_F(x, x') \approx f_1(x) - f_1(x')$ for a pair differing only on subset $F$ , instead of absolute regression from the full encoding (as in NSGANetV2)                                             | This is the property that separates P3Net from NSGANetV2 already at the problem-formulation level                                                                  | Problem Formulation              |
| Mechanism lineage                   | Mechanistically closer to CS-GOMEA (regression on continuous fitness differences, no formal guarantee); philosophically closer to eLyMPuS (relative estimator over an incompletely discovered dependency structure, not assumed known in advance) | NAS validation accuracy does not decompose into subfunctions (no gray-box access); P3Net accordingly forgoes eLyMPuS's monotonicity-conditional recovery guarantee | Related Work; Proposed Optimizer |
| Reconstructing an absolute estimate | Telescoping along the chain of accepted modifications back to the nearest fully-evaluated ancestor: $\hat{f}_1(x_m) = f_1(x_0) - \sum_i \hat{\delta}_{F_i}$                                                                                       | Scores a candidate deep in the sweep without a full evaluation at every step                                                                                       | Proposed Optimizer               |
| Ancestor guarantee                  | $x_0$ is always drawn from $\mathcal{H}_t$ (completed steps 5–6), never a surrogate-only individual mid-sweep                                                                                                                                     | Closes a gap where the telescoping construction could otherwise bottom out on an unknown $f_1(x_0)$                                                                | Proposed Optimizer               |
| Donor provenance (open)             | Whether the donor drawn in step 1 must also come from $\mathcal{H}_t$ , like the parent and $x_0$ , or may be a transient surrogate-only individual, is not yet stated                                                                            | Flagged as TODO in text; parents and $x_0$ already carry this restriction, the donor does not yet                                                                  | Proposed Optimizer               |
| Surrogate training                  | On pairwise differences drawn from $\mathcal{H}_t$ (the set of full evaluations so far)                                                                                                                                                           | —                                                                                                                                                                  | Problem Formulation              |
| Refresh cycle                       | $\mathcal{H}_t$ , $\hat{\delta}_F$ , and the linkage tree are rebuilt after every round of full evaluations                                                                                                                                       | Surrogate and dependency structure co-evolve with the population                                                                                                   | Proposed Optimizer               |

### Search loop

| Step | What happens                                                                                      | Location           |
|------|---------------------------------------------------------------------------------------------------|--------------------|
| 1    | P3 generates candidates through a full optimal-mixing sweep along the linkage tree                | Proposed Optimizer |
| 2    | Every proposal is scored relative to its immediate predecessor in the sweep, via $\hat{\delta}_F$ | Proposed Optimizer |

| Step | What happens                                                                                                                                                           | Location           |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 3    | Tentative acceptance if the surrogate predicts non-negative improvement (zero threshold by default, swept in Results); otherwise rejection and move to the next subset | Proposed Optimizer |
| 4    | Once the sweep completes (or is cut short by $\kappa$ ), $C^*$ is selected as the candidates nondominated in $(\hat{f}_1, f_2)$                                        | Proposed Optimizer |
| 5    | Full evaluation of every $x \in C^*$                                                                                                                                   | Proposed Optimizer |
| 6    | Update $\mathcal{H}_t$ , retrain $\hat{\delta}_F$ , rebuild the linkage tree                                                                                           | Proposed Optimizer |

#### *Safeguards and auxiliary parameters*

| Decision                            | Specifics                                                                                                                                                                                                                                                                                                                                               | Rationale / consequence                                                                                                                                                                                            | Location                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Deduplication                       | Candidates already present in $\mathcal{H}_t$ skip re-evaluation; applied identically to every compared method as part of the shared evaluation harness, not just within P3Net’s loop                                                                                                                                                                   | Direct response to the genotype-to-phenotype redundancy documented in NSGA-Net; uniform application prevents the saving from inflating P3Net’s apparent efficiency relative to baselines that could equally use it | Proposed Optimizer; Results, Fairness controls |
| Chain depth $\kappa$                | Maximum number of tentatively accepted modifications before a full evaluation is forced                                                                                                                                                                                                                                                                 | Bounds the accumulation of surrogate error along a sweep                                                                                                                                                           | Proposed Optimizer                             |
| Acceptance threshold                | Step-3 acceptance rule uses a zero threshold by default ( $\hat{\delta}_F \geq 0$ )                                                                                                                                                                                                                                                                     | Simplest heuristic consistent with standard optimal-mixing acceptance (no worsening); swept jointly with $\kappa$ rather than assumed adequate                                                                     | Proposed Optimizer; Results                    |
| Default value of $\kappa$           | Initialised at eLyMPuS’s $2\lceil \log_2 n \rceil$ -step bound, where $n$ is the dimensionality of the full extended, discretised genotype (architecture + $\Theta$ ), not the architecture-only count of Problem Formulation — as a heuristic, <i>not</i> a derivation from the monotonicity guarantee that bound relies on (which does not hold here) | Honest acknowledgement that the theoretical guarantee does not transfer to NAS; sensitivity tested empirically, jointly with the acceptance threshold above                                                        | Proposed Optimizer; Results                    |
| Full sweep, not single modification | Candidate generation always runs the complete optimal-mixing sweep                                                                                                                                                                                                                                                                                      | Consistency with the variation operator P3’s own guarantees rely on                                                                                                                                                | Proposed Optimizer                             |
| No guarantee for $\hat{\delta}_F$   | Treated as a learned heuristic estimator, without eLyMPuS’s recovery guarantee (monotonicity assumption broken for NN fitness)                                                                                                                                                                                                                          | Scientific caution — does not import a theoretical guarantee into a domain where it does not hold                                                                                                                  | Proposed Optimizer                             |
| Nondomination reference frame       | Step 4 compares only candidates from the same search iteration (same $\mathcal{H}_t$ snapshot)                                                                                                                                                                                                                                                          | Prevents mixing $\hat{f}_1$ estimates against a drifting surrogate across iterations                                                                                                                               | Proposed Optimizer                             |

#### *Benchmarks and experimental scope*

| Decision                            | Specifics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rationale / consequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Location              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Primary benchmarks                  | JAHS-Bench-201 (Category 2 — continuous surrogate, no enumerable oracle) and NAS-HPO-Bench-II (Category 1 — discrete grid, exact oracle) — both, deliberately, side by side                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Deterministic objective, computable oracle front, bounded evaluation budget; a result holding on both is reported as primary, on one as benchmark-specific                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Related Work; Results |
| Baselines                           | NSGANetV2 (primary, discretised $\Theta$ ); NSGANetV2 extended to the joint setting with an unconstrained, real-valued $\Theta$ (control, isolating the effect of discretisation itself); NSGA-Net with no surrogate, every candidate fully evaluated (engine-only ablation, NSGA-II side, symmetric to the P3 row below); P3 without a surrogate (engine-only ablation, P3 side); P3 with an absolute regressor in the NSGANetV2 style (surrogate-only ablation); SH-EMOA and MO-BOHB, the baselines established for this exact joint setting on these same two benchmarks [Guerrero-Viu et al., 2021]; random search and TPE (sanity checks) | Attributes gains to search engine vs. surrogate vs. discretisation as far as the design permits; not a full orthogonal factorial, since the relative surrogate $\hat{\delta}_F$ is only defined given a linkage tree, so an NSGA-II-plus-relative-surrogate cell does not exist. SH-EMOA/MO-BOHB added because, unlike NSGANetV2, they were never imported from an architecture-only setting and require no such extension                                                                                                                                                                                                      | Results               |
| P3-without-surrogate budget reality | Follows the canonical optimal-mixing acceptance rule: every accepted step within a sweep is gated by a <i>real</i> $f_1$ evaluation, as in P3’s own hill-climbing/cross-level mixing                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Expected to exhaust most or all of a 50–200 budget within very few sweeps; reported explicitly (number of full sweeps completed within budget) so a weak result reads as budget starvation, not engine failure — this collapse is itself informative, motivating the surrogate                                                                                                                                                                                                                                                                                                                                                  | Results, Baselines    |
| Full evaluation budget tiers        | Three tiers, $\{50, 100, 200\}$ full evaluations, doubling at each step; $R = 10$ independent seeds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Verified against three literature-specific figures, not chosen arbitrarily: (1) JAHS-Bench-201’s own suggested protocol is $\approx 100$ evaluations, min. 10 seeds [Bansal et al., 2022] – matches the middle tier and $R$ directly; (2) NSGANetV2 itself uses exactly 350 full evaluations (Table 2) to reach its reported results [Lu et al., 2020] – the top tier (200) stays below this; (3) NAS-HPO-Bench-II’s own paper benchmarks search algorithms at 500 trials [Hirose et al., 2021] – all three tiers stay below this too, deliberately, since the central question is about a <i>limited</i> budget (Introduction) | Results               |
| Metrics                             | Fixed-budget hypervolume always; IGD+ only where the oracle front is enumerable (i.e. NAS-HPO-Bench-II only), otherwise hypervolume relative to a best-known front (open: precise definition, e.g. union of all points evaluated by any compared method, still TODO)                                                                                                                                                                                                                                                                                                                                                                           | Metric matched to what each benchmark actually offers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Results               |

| Decision                      | Specifics                                                                                                                                                                                                      | Rationale / consequence                                                                                                                                                                                                                                     | Location                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Statistical plan              | Paired Wilcoxon signed-rank test, Holm–Bonferroni correction, effect size (e.g. Cliff’s delta) reported with every corrected $p$ -value                                                                        | Deliberately stricter than the uncorrected comparisons in Bartnik                                                                                                                                                                                           | Results                   |
| Comparison set for correction | P3Net vs. each of the other nine arms, per benchmark and per budget tier — not every pairwise combination among all arms                                                                                       | Makes explicit what Holm–Bonferroni is correcting across, since the corrected significance threshold depends on this count                                                                                                                                  | Results, Statistical plan |
| Diagnostics                   | Genotype duplication rate and archive turnover, measured under identical genotype encoding <i>and</i> an identically applied deduplication cache across methods; counted at proposal time, not evaluation time | Empirically tests whether the NSGA-Net-derived motivation (duplication redundancy) actually transfers to the joint arch+HP genotype, apples-to-apples across methods, without the comparison being confounded by asymmetric caching or by measurement point | Results                   |
| Reproducibility               | Cache keyed by genotype, experiment type, and a protocol version string                                                                                                                                        | Automatic invalidation of stale cache entries on any settings change                                                                                                                                                                                        | Results                   |

*Scope boundaries (deliberately excluded / distinguished)*

| Decision                                      | Specifics                                                                                                                                                                            | Location     |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Gray-box vs. black-box                        | P3Net operates in the black-box regime (NAS accuracy is not decomposable); explicitly distinguished from the gray-box optimisation of Whitley/Chicano/Goldman used in the GOMEA line | Related Work |
| Benchmark surrogate vs. search-time surrogate | JAHS-Bench-201’s surrogate stands in for the objective once, offline, before search begins; P3Net’s surrogate is fitted incrementally during search — two independent design choices | Related Work |
| Delta vs. closest prior work (Bartnik)        | Bartnik: absolute regressor, MO-LS/MO-P3-GOMEA baselines (not NSGA-II/NSGANetV2), architecture-only search (NAS-Bench-201, no $\Theta$ ) — P3Net differs on all three axes at once   | Related Work |

*Settled review decisions (do not re-open without new information)*

Before flagging an issue in a new review pass, check this table first. “Fixed” items have already been addressed in the text at the cited location. “Rejected” items were considered and deliberately left unchanged – re-raising them as if undiscovered is not useful without a concrete reason the rejection no longer holds. This table itself is exempt from review, per the standard reviewer instructions.

| Item                                                                   | Status   | Resolution / reason                                                                                                                                          | Where                                |
|------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Related Work placed after Proposed Optimizer                           | Rejected | Author’s explicit choice, confirmed after review; forward references (eLyMPuS, GOMEA lineage) are glossed inline at first use instead of reordering sections | Introduction; Proposed Optimizer     |
| Deduplication cache scope                                              | Fixed    | Applies identically to every compared method as shared harness, not only P3Net                                                                               | Proposed Optimizer; Results          |
| Discretisation asymmetry (NSGA-II loses native crossover, P3 does not) | Fixed    | Bounded empirically via a native-continuous- $\Theta$ NSGANetV2 control baseline, not just acknowledged                                                      | Results, Baselines/Fairness controls |
| Engine $\times$ surrogate ablation not a full factorial                | Fixed    | Stated explicitly: relative surrogate needs a linkage tree, so an NSGA-II+relative-surrogate cell cannot exist                                               | Results, Baselines                   |

| Item                                                                                          | Status             | Resolution / reason                                                                                                                                                                                                           | Where                                            |
|-----------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| $\mathcal{H}_t$ vs. $D$ notational collision                                                  | Fixed              | Observation dataset renamed $\mathcal{H}_t$ throughout                                                                                                                                                                        | Problem Formulation; Proposed Optimizer; Results |
| Multi-fidelity ladder apparatus vs. unused in practice                                        | Fixed              | Stated explicitly that no current baseline exploits it; kept as general infrastructure for a possible future multi-fidelity baseline                                                                                          | Problem Formulation                              |
| Full evaluation budget tiers unsourced                                                        | Fixed              | $\{50, 100, 200\}$ , $R = 10$ , each anchored to a verified literature figure (JAHS-Bench-201 protocol, NSGANetV2's 350, NAS-HPO-Bench-II's 500)                                                                              | Results, Budgets                                 |
| "P3 alone" ablation: acceptance mechanism and budget feasibility undefined                    | Fixed              | Canonical optimal-mixing acceptance (real $f_1$ gates every accepted step) stated explicitly; expected budget starvation stated as an informative result, not hidden; sweeps-completed-within-budget reported as a diagnostic | Results, Baselines                               |
| Missing NSGA-II-side engine-only baseline (asymmetric with "P3 alone")                        | Fixed              | Added: NSGA-Net (no surrogate, full training every candidate) as the symmetric engine-only ablation                                                                                                                           | Results, Baselines                               |
| $s$ (evaluation-noise seeds) vs. $R$ (independent runs) terminology collision                 | Fixed              | Distinguished explicitly; both chosen benchmarks are deterministic, so $s = 1$ throughout                                                                                                                                     | Problem Formulation                              |
| $\kappa$ 's default bound: does $n$ include $\Theta$ ?                                        | Fixed              | Clarified: $n$ is the full extended, discretised genotype dimensionality                                                                                                                                                      | Proposed Optimizer                               |
| Telescoping construction: is the ancestor $x_0$ guaranteed fully evaluated?                   | Fixed              | Stated explicitly: parents and $x_0$ are always drawn from $\mathcal{H}_t$ , never surrogate-only individuals                                                                                                                 | Proposed Optimizer                               |
| Linkage tree rebuild granularity (within a sweep vs. between rounds)                          | Fixed              | Stated explicitly: fixed within an iteration, rebuilt only at step 6                                                                                                                                                          | Proposed Optimizer                               |
| Statistical plan: "full set of comparisons" undefined                                         | Fixed              | Enumerated: P3Net vs. each of the other nine arms, per benchmark and budget tier                                                                                                                                              | Results, Statistical plan                        |
| Diagnostics: duplication rate measured pre- or post-dedup-cache?                              | Fixed              | Stated explicitly: counted at proposal time, not evaluation time                                                                                                                                                              | Results, Diagnostics                             |
| P3 pyramid / "parameter-less" never explained                                                 | Fixed              | One paragraph added at first substantive mention of P3                                                                                                                                                                        | Related Work                                     |
| NSGANetV2's "few hundred" evaluations left vague while Results cites 350                      | Fixed              | Related Work now states the precise figure (350) at first mention                                                                                                                                                             | Related Work                                     |
| Budget-tier justification citing benchmark-mismatch as a reason for the dual-benchmark design | Fixed              | Circular clause removed; the dual-benchmark choice is justified once, in Related Work (Category 1/2), not re-derived from budget numbers                                                                                      | Results, Budgets                                 |
| "in the spirit of eLyMPuS" phrase repeated across three sections                              | Fixed (superseded) | No longer applies: the Proposed Optimizer instance was replaced by the CS-GOMEA/eLyMPuS mechanism-vs-philosophy split above, so the phrase now appears once, in Introduction, as a forward-reference gloss                    | Introduction                                     |
| Three-surveys citation appearing twice in Related Work                                        | Rejected           | Not a duplication: the two passages draw two distinct claims (dominant paradigm; absence of linkage learning as a category) from the same three surveys                                                                       | Related Work                                     |
| No explicit directional hypothesis (H1)                                                       | Rejected           | Kept as a neutral research question by deliberate choice; GECCO framing convention, not a defect                                                                                                                              | Introduction                                     |

| Item                                                                                                                                     | Status                | Resolution / reason                                                                                                                                                                                                                                                                                                                                                    | Where                                     |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Results register (“must not allow” imperative phrasing)                                                                                  | Deferred              | Results is still a protocol draft; register will be normalised in the final copy-edit pass once real numbers replace the sketch, not iteratively now                                                                                                                                                                                                                   | Results                                   |
| “sanity baselines” wording                                                                                                               | Deferred              | Low-value terminology preference; bundle into the final copy-edit pass                                                                                                                                                                                                                                                                                                 | Results, Baselines                        |
| Introduction foreshadowing the duplication-rate diagnostic                                                                               | Fixed (opportunistic) | Folded into the same edit that hedged the duplicate-vs-interaction correlation claim, since both touched the same sentence                                                                                                                                                                                                                                             | Introduction                              |
| NAS-HPO-Bench-II: which fidelity level constitutes $r_K$ (tabulated 12 epochs vs. 200-epoch GIN+MLP surrogate)                           | Fixed                 | Confirmed: $r_K$ is fixed at the tabulated range only; the 200-epoch surrogate extrapolation is never queried, so the Category 1 classification and exact-oracle-front status both hold                                                                                                                                                                                | Related Work; Results                     |
| “Analogous to eLyMPuS” surrogate framing conflated mechanism with philosophy                                                             | Fixed                 | Split explicitly: mechanism (regression on continuous differences, no guarantee) is closer to CS-GOMEA; philosophy (relative estimator over an incompletely discovered structure) is closer to eLyMPuS                                                                                                                                                                 | Related Work; Proposed Optimizer          |
| JAHS-Bench-201 search space reported as 14-dimensional, conflating the search space with search+fidelity                                 | Fixed                 | Corrected to ten dimensions for the search space alone (six architecture edges plus four hyperparameters); fourteen only once JAHS-Bench-201’s own four fidelity dimensions are added, per the source’s own figure                                                                                                                                                     | Results                                   |
| No explicit statement that P3Net has no mutation operator                                                                                | Fixed                 | Added: all variation is optimal mixing, inherited from the P3/GOMEA lineage                                                                                                                                                                                                                                                                                            | Proposed Optimizer                        |
| Donor provenance in step 1: restricted to $\mathcal{H}_t$ like the parent and $x_0$ , or may be transient?                               | Open                  | Not yet resolved; flagged as TODO in text, requires an authorial decision                                                                                                                                                                                                                                                                                              | Proposed Optimizer                        |
| Best-known-front definition for the hypervolume fallback on non-enumerable benchmarks                                                    | Open                  | Not yet resolved; flagged as TODO in text, candidate definition is the union of all points evaluated by any compared method across all runs                                                                                                                                                                                                                            | Results, Metrics                          |
| JAHS-Bench-201 surrogate approximation error never acknowledged as a limitation                                                          | Fixed                 | Added: since every compared method queries the same surrogate, systematic bias affects them equally, but this does not rule out the error correlating specifically with the structure P3Net exploits; a result that also holds on NAS-HPO-Bench-II (measured data) is therefore treated as the primary finding, a JAHS-Bench-201-only result as requiring confirmation | Conclusions                               |
| Why benchmarks (rather than live training) are used at all, left unstated                                                                | Fixed                 | Added: same rationale as NAS-Bench-101 [Ying et al., 2019] — precomputing or approximating the objective once lets many search runs be compared on identical, reproducible ground truth instead of repeating live training for every candidate and every compared method                                                                                               | Related Work                              |
| Polish “eLyMPuS, wariant LyMPuS” phrasing readable as backwards (undeclined foreign acronym leaves the genitive-like relation ambiguous) | Fixed                 | Reworded to “eLyMPuS jest wariantem LyMPuS” (instrumental case), which fixes the direction unambiguously; the English “a variant of LyMPuS” was already unambiguous via “of” and needed no change                                                                                                                                                                      | Introduction                              |
| Paper had only one diagram (the P3Net search-loop figure); reviewer-style question of whether more would help readability                | Fixed                 | Added three further diagrams: an engine×surrogate ablation grid (table), a P3 population-pyramid figure, and an illustrative linkage-tree/subset- $F$ figure; every LaTeX figure has a corresponding Markdown rendering (ASCII/table) rather than a bare cross-reference                                                                                               | Results; Related Work; Proposed Optimizer |

| Item                                                                                                                                                                                                                                                                                                                                                | Status | Resolution / reason                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Where                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| External reviewer flagged citation hygiene as “messy” / “not by the book”                                                                                                                                                                                                                                                                           | Fixed  | Removed one redundant back-to-back re-citation of the same source within a single paragraph (P3 pyramid, [Goldman and Punch, 2014]); normalised every citation to a non-breaking space ( <code>\citep{}</code> ), 33 instances previously used a plain space, risking an orphaned citation marker at a line break); replaced 4 bare <code>\cite{}</code> instances with <code>\citep{}</code> for consistency, since the paper otherwise uses <code>\citep</code> exclusively and never <code>\citet</code>                                                                                                                                                                                                                                                                                        | Related Work; Results      |
| Self-inflicted regression from the citation-spacing fix above: the automated substitution collided with existing <code>\</code> (control-space) macros after “et al.”, corrupting 5 instances into <code>\~\citep{}</code> (a stray backslash before the tilde, which LaTeX would parse as the combining-tilde-accent command, not a spacing tilde) | Fixed  | Caught while auditing citations for the Word-export pipeline; corrected all 5 instances ( <code>et al.\ \citep → et al.\~\citep</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Introduction; Related Work |
| External reviewer (via annotated PDF, <code>NAS_com_v1.pdf</code> ) flagged three further citation issues                                                                                                                                                                                                                                           | Fixed  | (1) Introduction’s opening two sentences (NAS’s dominant cost; sequential architecture/HP tuning as common practice) were unsourced before the first citation three sentences in; grounded each in existing citations ([White et al., 2023, Liu et al., 2023] and [Lu et al., 2019, 2020] respectively). (2) Three instances spelled out all authors of a 3+-author work in running prose instead of “et al.” (Whitley/Chicano/Goldman; Dushatskiy/Alderliesten/Bosman; Tran/Truong/Vo/Luong); shortened to “et al.” throughout, matching the convention used everywhere else in the paper. (3) The gap-analysis paragraph named “Bartnik” three times with no citation attached to any of the three (nearest [Bartnik, 2026] was two paragraphs earlier); added the citation at each bare mention | Introduction; Related Work |